

Multiple comparisons

Project: Reproducible AI Research Study

1. Your mission

Your learning target

I can explain the mechanism and terminology of multiple comparisons.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of multiple comparisons

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Statistical tests and assumptions. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain multiple comparisons using the records and boundaries of this fictional investigation.

2. Notice the evidence

Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

For this exercise use Bonferroni: family alpha .05 divided by number of tests m ; reject each test when its valid p-value is at most $.05/m$. This controls the family error by a bound without requiring independent tests.

Multiple comparisons — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Five tests have p-values .009, .012, .3, .4, .8.

Check or calculation: For this exercise use Bonferroni: family alpha .05 divided by number of tests m ; reject each test when its valid p-value is at most $.05/m$. This controls the family error by a bound without requiring independent tests.

Conclusion and limit: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Five tests have p-values .009, .012, .3, .4, .8. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: The declared family has ten tests instead of six; keep the listed values unchanged.

Prediction: The cutoff becomes 0.005, and only 0.004 among the listed values still rejects.

Reason: The error allowance is divided among ten tests; all ten results must be reported, including the four not supplied here.

5. Words that unlock the idea

Word · Meaning

Word: Multiple comparisons

Meaning: Testing several hypotheses in one declared family

Word: Family error

Meaning: The chance of at least one false rejection under relevant nulls

Word: Bonferroni rule

Meaning: Dividing a family threshold by the number of tests

Use two terms to explain a step in this week's example.

6. Follow a worked example

Five tests have p-values .009, .012, .3, .4, .8.

Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
def bonferroni(p_values, alpha=0.05):  
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in  
p_values)):  
        raise ValueError('test settings')  
    threshold = alpha / len(p_values)  
    return (threshold, [p <= threshold for p in p_values])  
  
print(bonferroni([.009, .012, .3, .4, .8]))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Multiple comparisons

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

The facts we will use

Familywise $\alpha=0.05$. Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

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Supplied p-values and six-test decisions

	Supplied p	At cutoff 0.00833
Test 1	0.004	Reject
Test 2	0.012	Do not reject
Test 3	0.02	Do not reject
Test 4	0.08	Do not reject
Test 5	0.4	Do not reject
Test 6	0.9	Do not reject

Read a cell using both its row and column labels.

Multiple comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count the planned tests

There are six tests in the declared family.



2. Compute corrected threshold

$0.05/6$ is about 0.00833 for each test.



3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Count the planned tests

What happens and why: There are six tests in the declared family.

Step: 2. Compute corrected threshold

What happens and why: $0.05/6$ is about 0.00833 for each test.

Step: 3. Apply all six decisions

What happens and why: Only 0.004 is below 0.00833; the other five do not reject under this rule.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Searching many comparisons provides more chances for false positives. Bonferroni divides the error allowance across the declared family; valid individual p-values are required. A rejected null is not automatically a large or useful improvement.

What this example does not establish

No effect size or real data is supplied. Bonferroni can be conservative and cannot repair invalid p-values or undisclosed outcome-driven test selection.

7. Find and repair a mistake

Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Multiple comparisons”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Multiple comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Five tests have p-values .009, .012, .3, .4, .8. Show the intermediate steps and give the conclusion with its scope.

```
def bonferroni(p_values, alpha=0.05):  
    if not p_values or not 0 < alpha < 1 or any((not 0 <= p <= 1 for p in  
p_values)):  
        raise ValueError('test settings')  
    threshold = alpha / len(p_values)  
    return (threshold, [p <= threshold for p in p_values])  
  
print(bonferroni([.009, .012, .3, .4, .8]))
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected

outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled multiple comparisons evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of multiple comparisons. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge

or reproduce it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Multiple comparisons — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Five tests have p-values .009, .012, .3, .4, .8.

Check or calculation: For this exercise use Bonferroni:family alpha .05 divided by number of tests m ;

reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.

Conclusion and limit: Threshold .01; only .009 rejected. An unadjusted .05 rule would also reject .012 and change family control.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Four tests have p-values .01, .02, .2, .9. Compute threshold and decisions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Four tests have p-values .01, .02, .2, .9. Compute threshold and decisions.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.